

NODAL ANALYSIS AND SUPERNODE TECHNIQUE



Lecture 6
First Class
Basics of Electrical Engineering

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NODAL ANALYSIS: THE CONCEPT

- We begin our study of general methods for methodical circuit analysis by considering a powerful method based on KCL, namely ***nodal analysis***.
- Every circuit has n nodes with one of the nodes being designated as a **reference (ground) node**.
- We designate the remaining $n - 1$ nodes as voltage nodes and give each node a unique name, V_i .
- At each node we write **Kirchhoff's current law** in terms of the node voltages.
- We form $n-1$ linear equations at the $n-1$ nodes in terms of the node voltages.
- We solve the $n-1$ equations for the $n-1$ node voltages.
- From the node voltages we can calculate any branch current or any voltage across any element.

STEPS OF NODAL ANALYSIS

1. Choose a reference (ground) node.

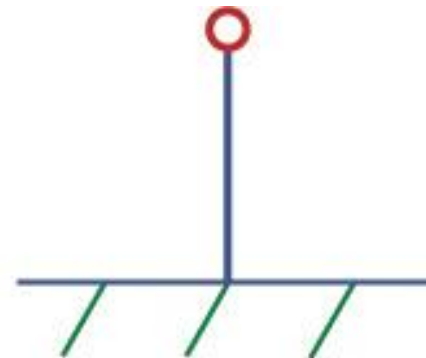
- The ground node is assumed to have 0 voltage.
- Common symbols for indicating a reference node, (a) common ground, (b) ground, (c) chassis.



(a)



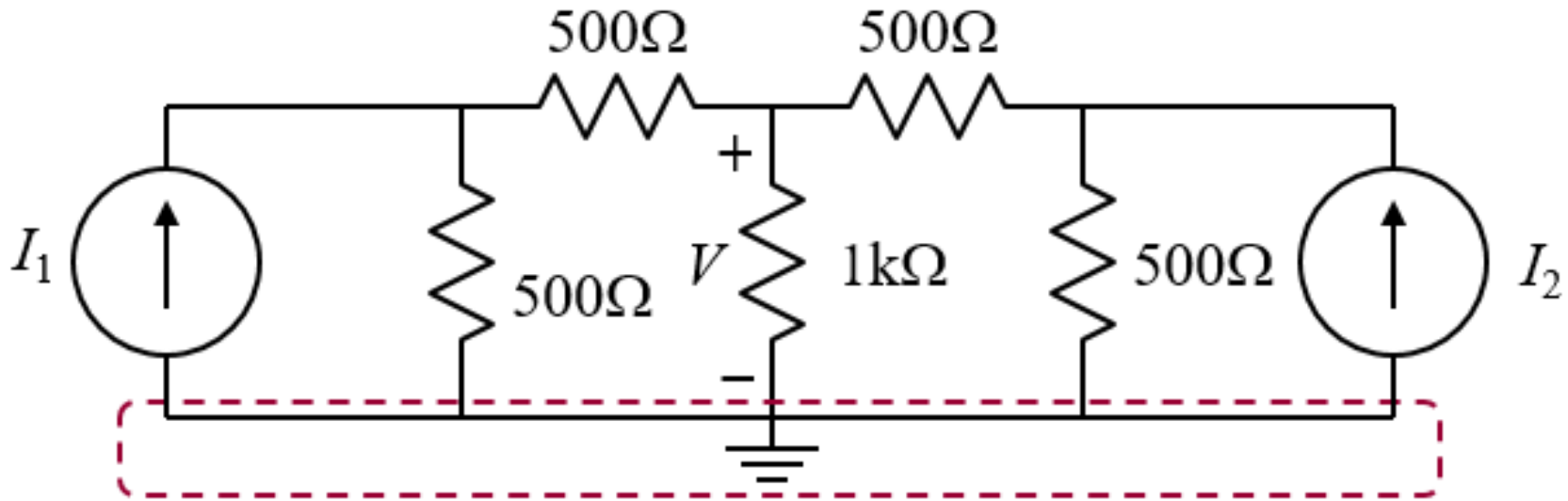
(b)



(c)

STEPS OF NODAL ANALYSIS

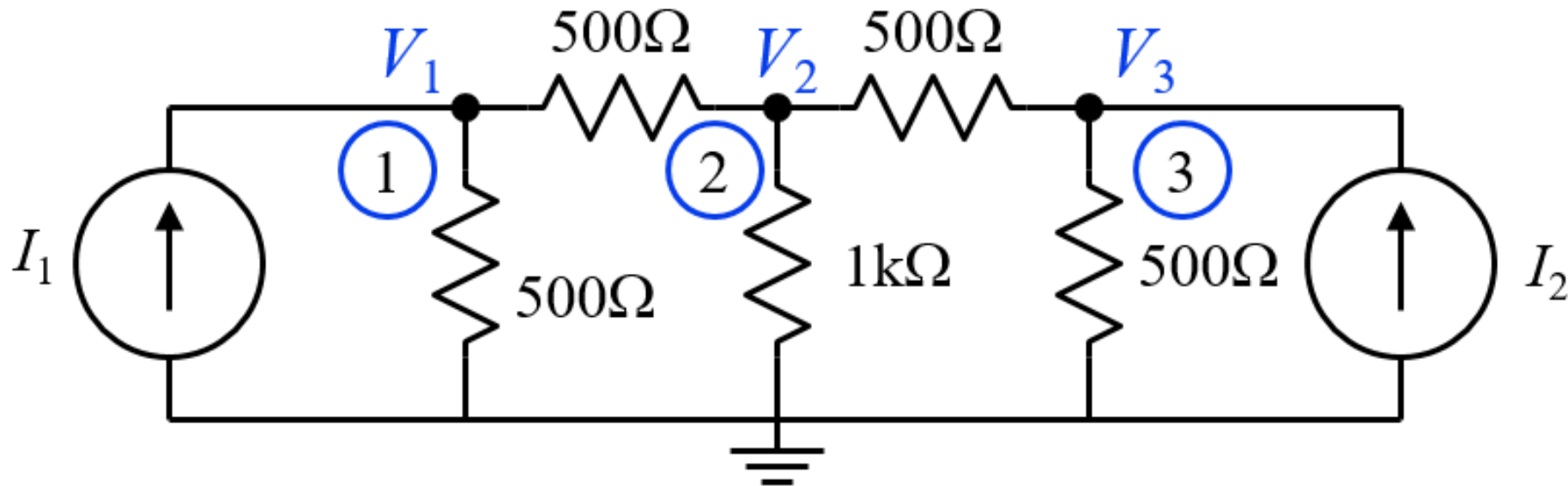
1. Choose a reference (ground) node.



The reference node is called the ground node
where $V = 0$

STEPS OF NODAL ANALYSIS

2. Assign node voltages to the other nodes.



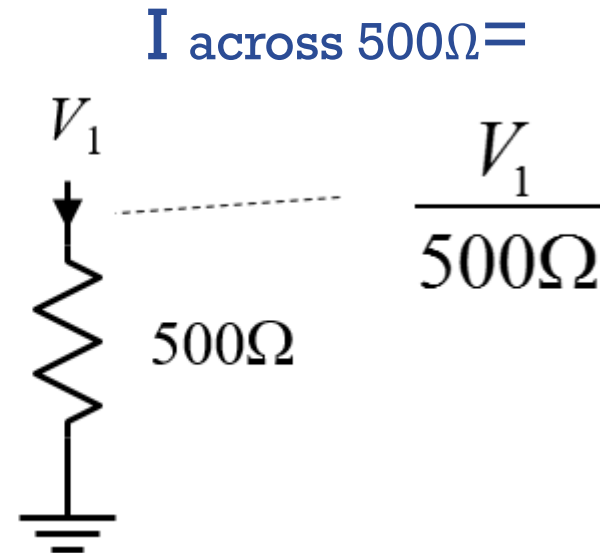
V_1 , V_2 , and V_3 are unknowns for which we solve using KCL

STEPS OF NODAL ANALYSIS

3. Apply KCL to each node other than the reference node;
express currents in terms of node voltages.

- Currents and Node Voltages

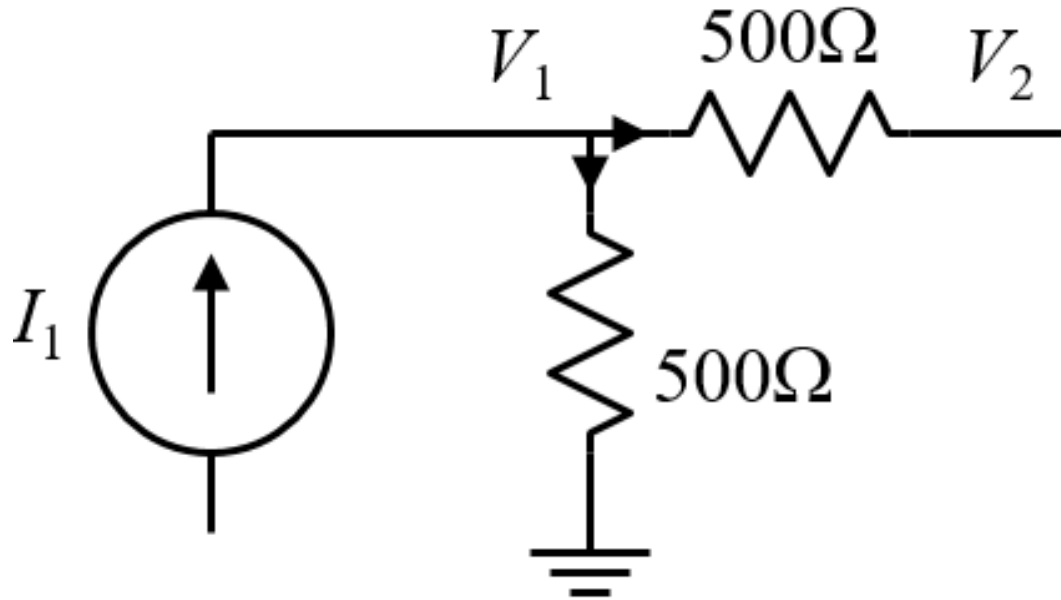

$$I_{\text{ across } 500\Omega} = \frac{V_1 - V_2}{500\Omega}$$


$$I_{\text{ across } 500\Omega} = \frac{V_1}{500\Omega}$$

STEPS OF NODAL ANALYSIS

3. Apply KCL to each node other than the reference node; express currents in terms of node voltages.

■ KCL at Node 1

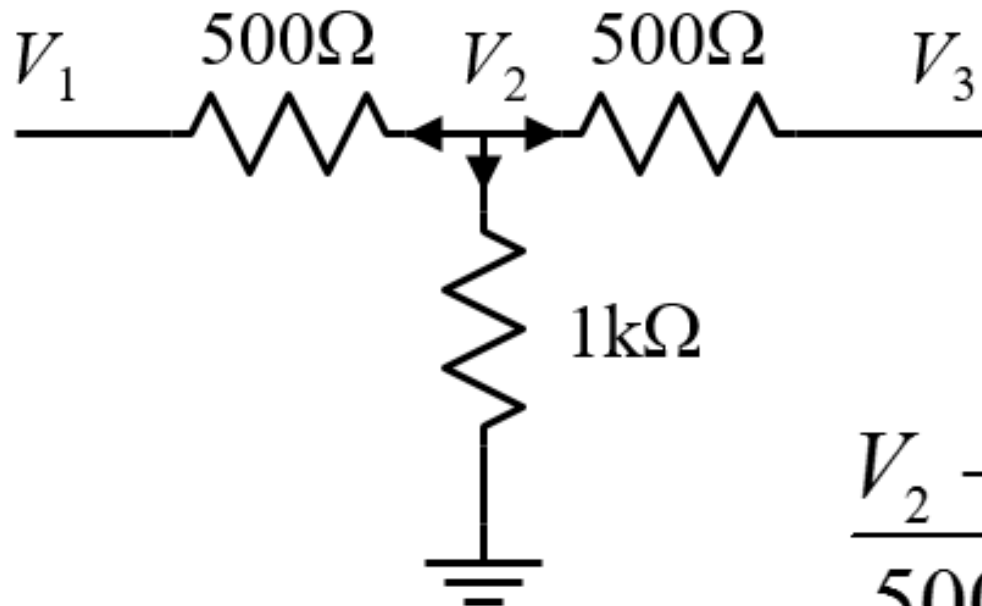


$$I_1 = \frac{V_1 - V_2}{500\Omega} + \frac{V_1}{500\Omega}$$

STEPS OF NODAL ANALYSIS

3. Apply KCL to each node other than the reference node; express currents in terms of node voltages.

■ KCL at Node 2

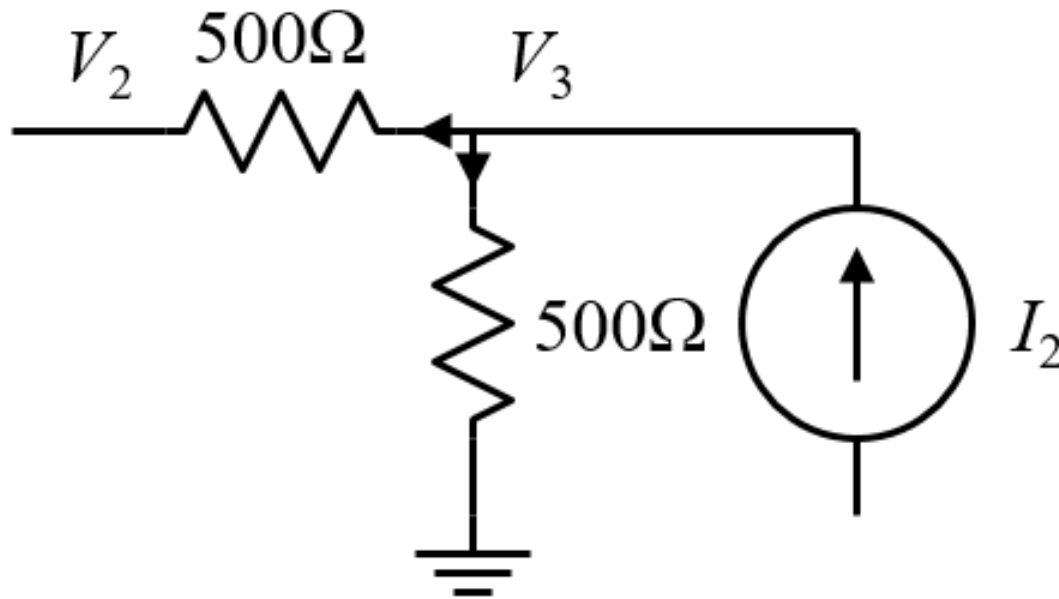


$$\frac{V_2 - V_1}{500\Omega} + \frac{V_2}{1k\Omega} + \frac{V_2 - V_3}{500\Omega} = 0$$

STEPS OF NODAL ANALYSIS

3. Apply KCL to each node other than the reference node; express currents in terms of node voltages.

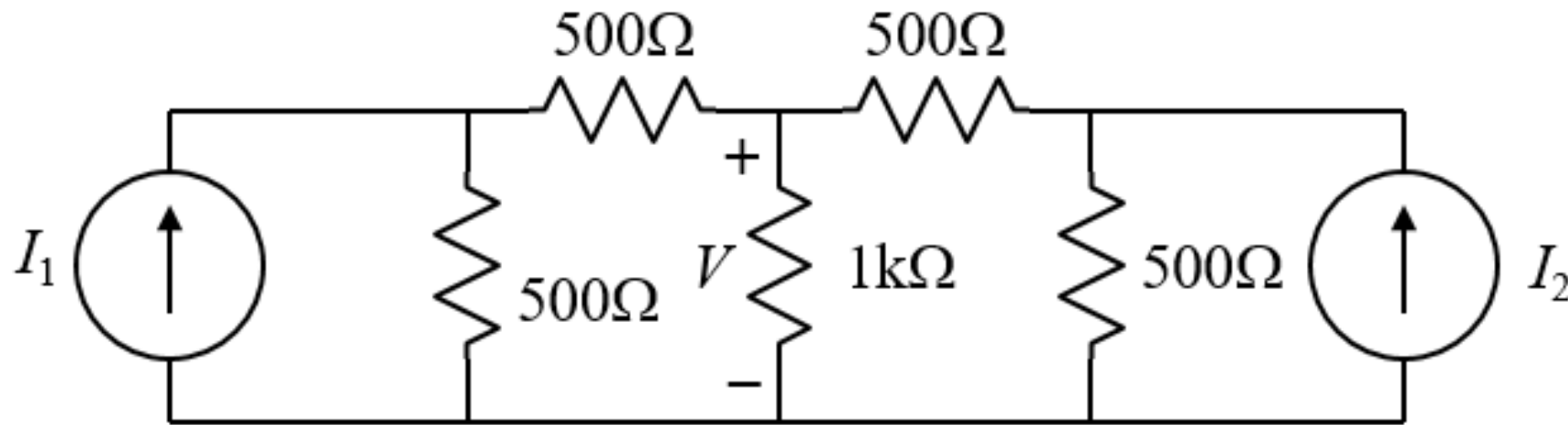
- KCL at Node 3



$$\frac{V_3 - V_2}{500\Omega} + \frac{V_3}{500\Omega} = I_2$$

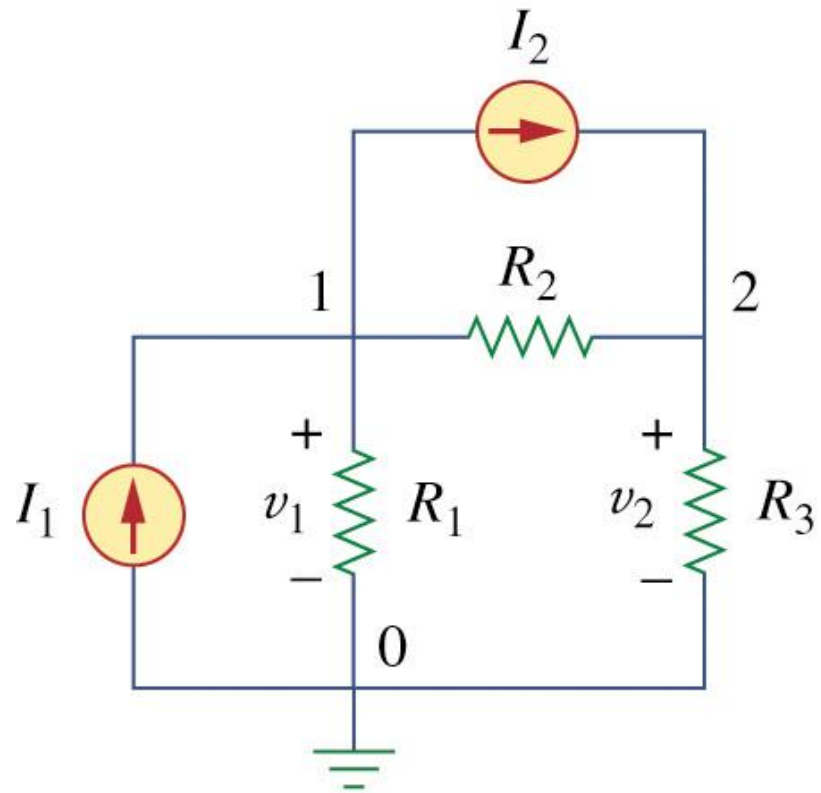
STEPS OF NODAL ANALYSIS

4. Solve the resulting system of linear equations for the nodal voltages.

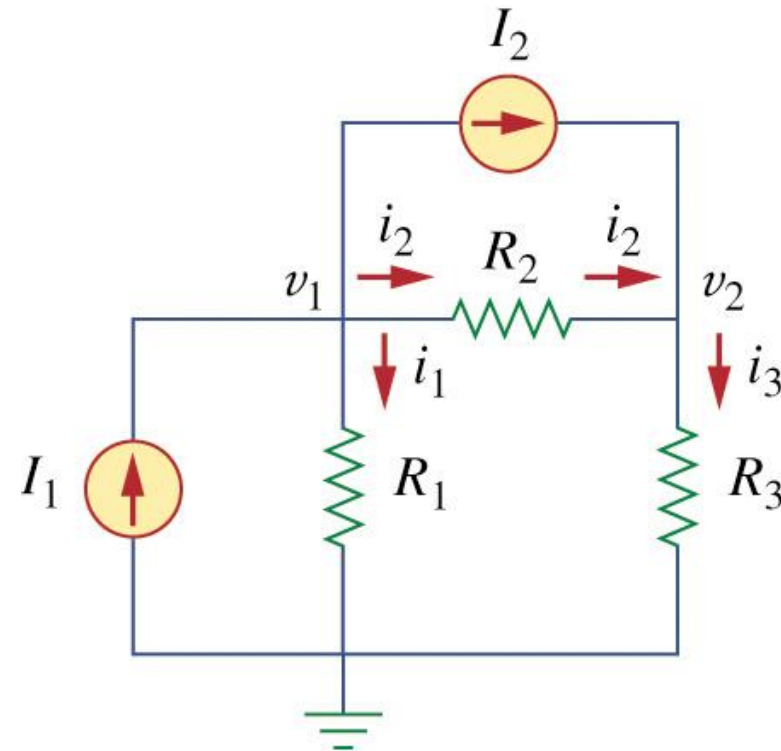


Solution: $V = 167I_1 + 167I_2$

TYPICAL CIRCUIT FOR NODAL ANALYSIS



(a)



(b)

TYPICAL CIRCUIT FOR NODAL ANALYSIS

$$I_1 = I_2 + i_1 + i_2$$

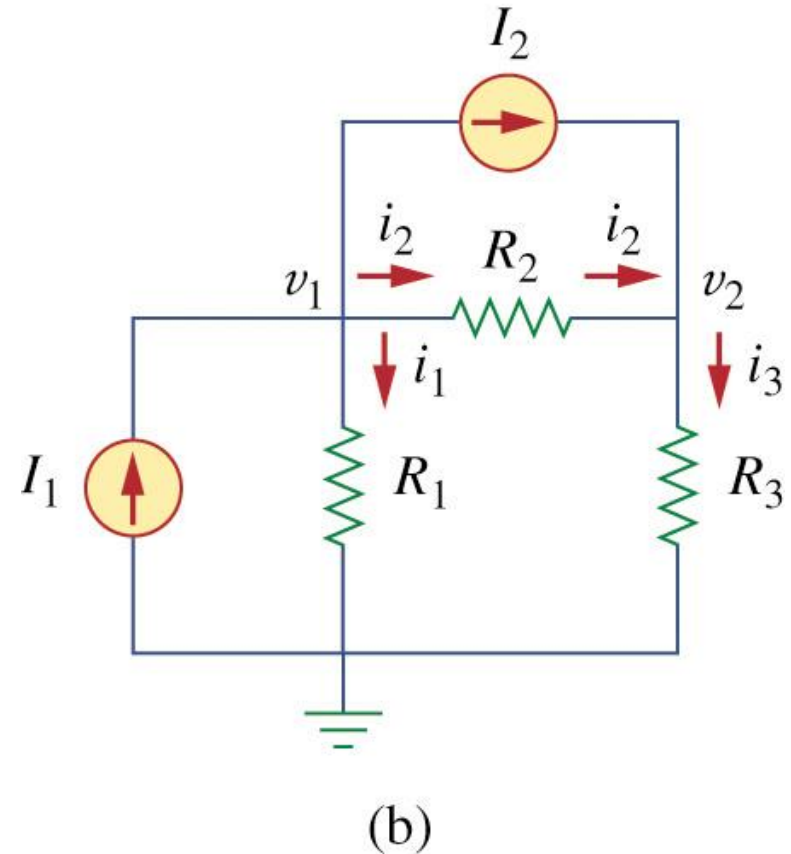
$$I_2 + i_2 = i_3$$

$$i = \frac{V_{\text{higher}} - V_{\text{lower}}}{R}$$

$$i_1 = \frac{v_1 - 0}{R_1} \quad \text{or} \quad i_1 = G_1 v_1$$

$$i_2 = \frac{v_1 - v_2}{R_2} \quad \text{or} \quad i_2 = G_2 (v_1 - v_2)$$

$$i_3 = \frac{v_2 - 0}{R_3} \quad \text{or} \quad i_3 = G_3 v_2$$



TYPICAL CIRCUIT FOR NODAL ANALYSIS

$$\Rightarrow I_1 = I_2 + \frac{v_1}{R_1} + \frac{v_1 - v_2}{R_2}$$

$$I_2 + \frac{v_1 - v_2}{R_2} = \frac{v_2}{R_3}$$

$$\Rightarrow I_1 - I_2 = G_1 v_1 + G_2 (v_1 - v_2)$$

$$I_2 = -G_2 (v_1 - v_2) + G_3 v_2$$

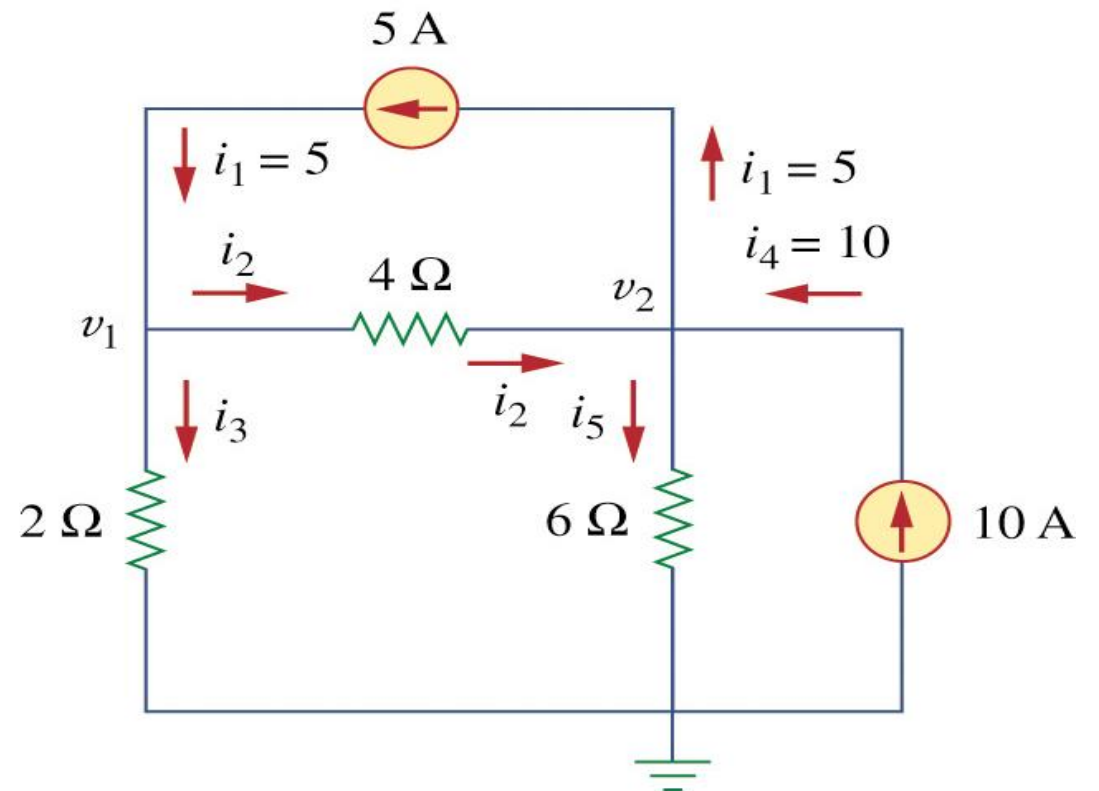
$$\Rightarrow \begin{bmatrix} G_1 + G_2 & -G_2 \\ -G_2 & G_2 + G_3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} I_1 - I_2 \\ I_2 \end{bmatrix}$$

EXAMPLE 1:

Calculate the node voltage in the circuit shown in Figure below:

- At node 1

$$i_1 = i_2 + i_3$$
$$\Rightarrow 5 = \frac{v_1 - v_2}{4} + \frac{v_1 - 0}{2}$$



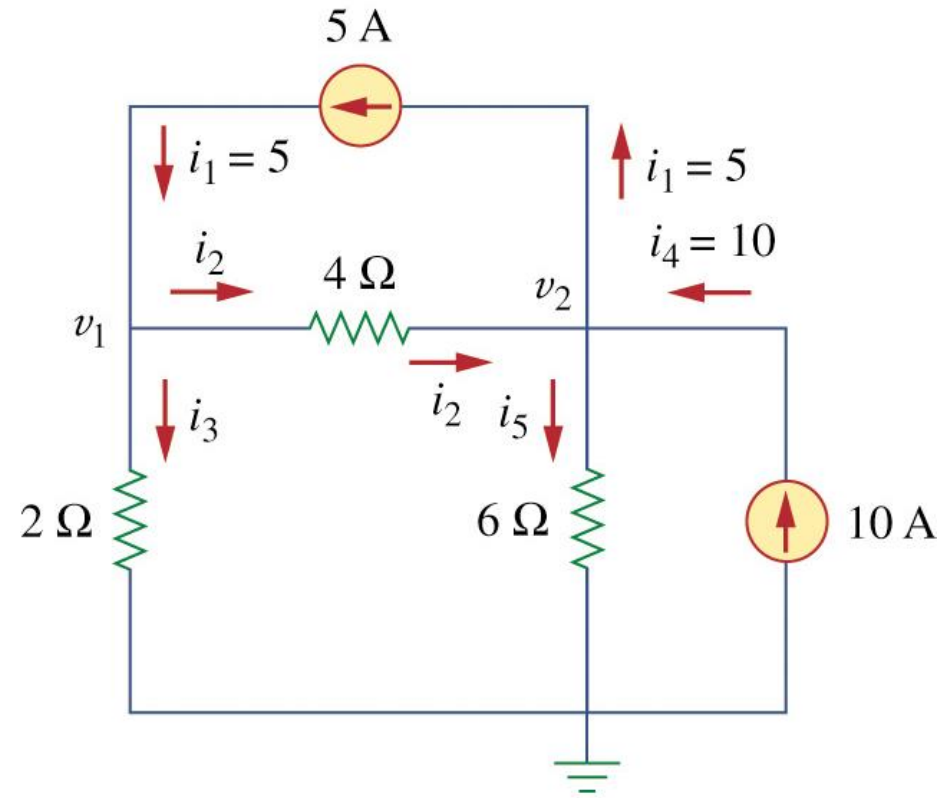
(b)

EXAMPLE 1:

- At node 2

$$i_2 + i_4 = i_1 + i_5$$

$$\frac{v_1 - v_2}{4} + 10 = 5 + \frac{v_2 - 0}{6}$$



(b)

EXAMPLE 1:

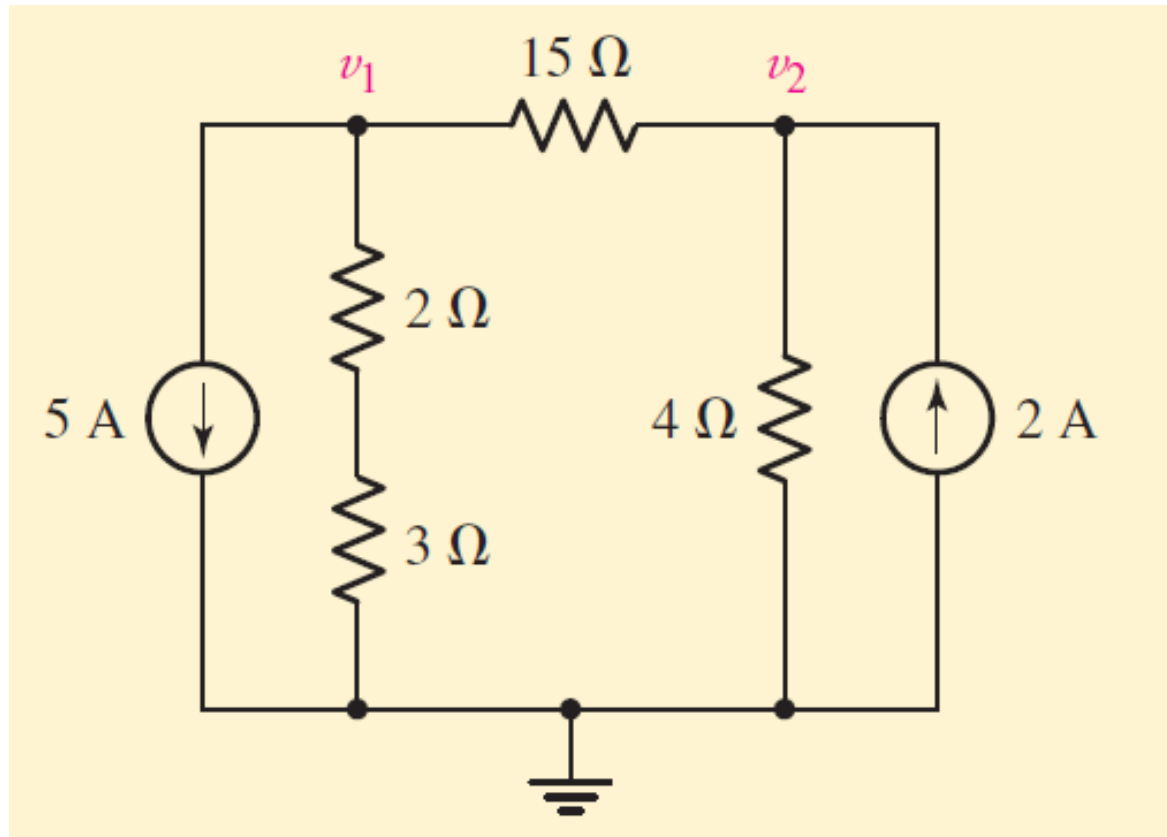
- In matrix form:

$$\begin{bmatrix} \frac{1}{2} + \frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{6} + \frac{1}{4} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} + \frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{6} + \frac{1}{4} \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

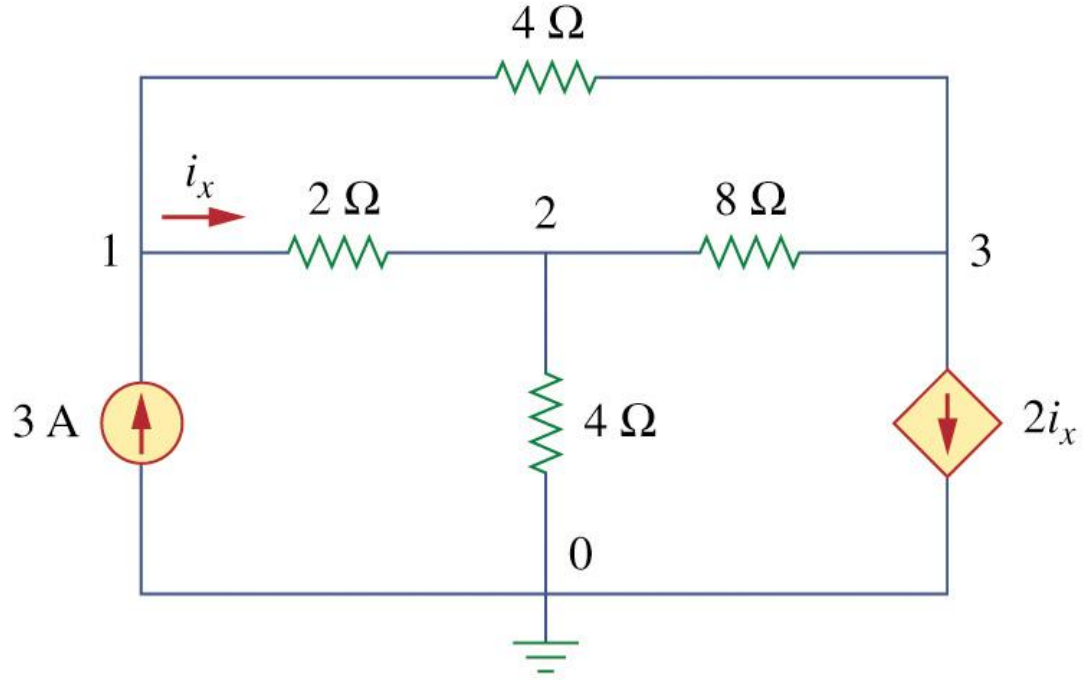
PRACTICE

For the circuit of Figure below, determine the nodal voltages v_1 and v_2 .

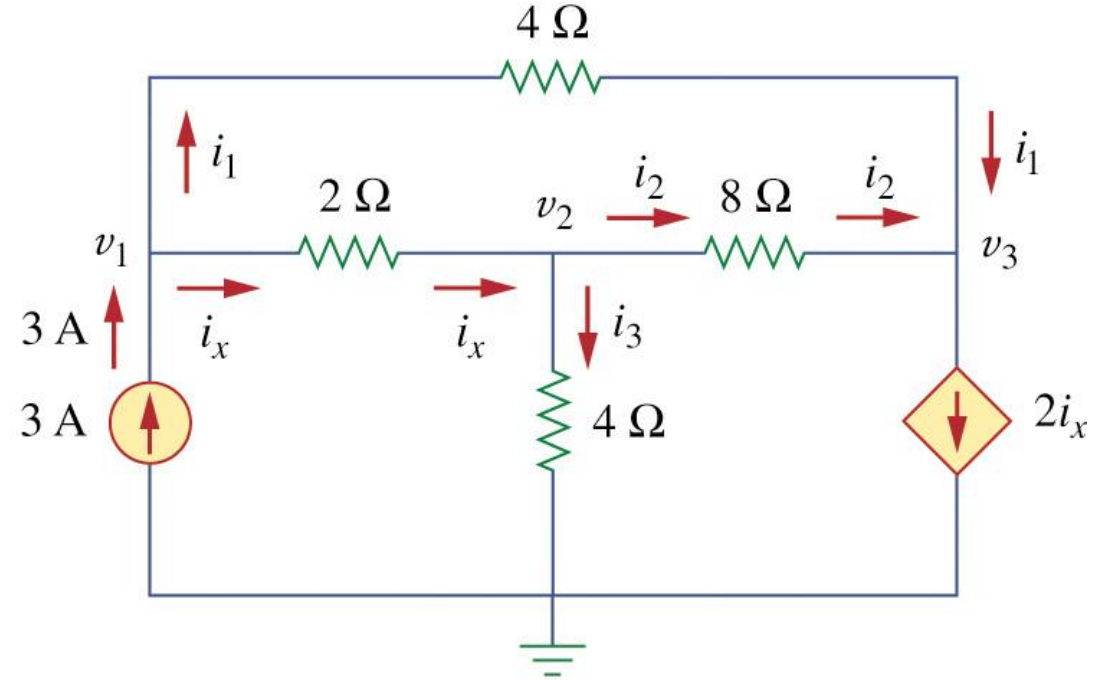


EXAMPLE 2:

Determine the voltage at the nodes in Figure below:



(a)

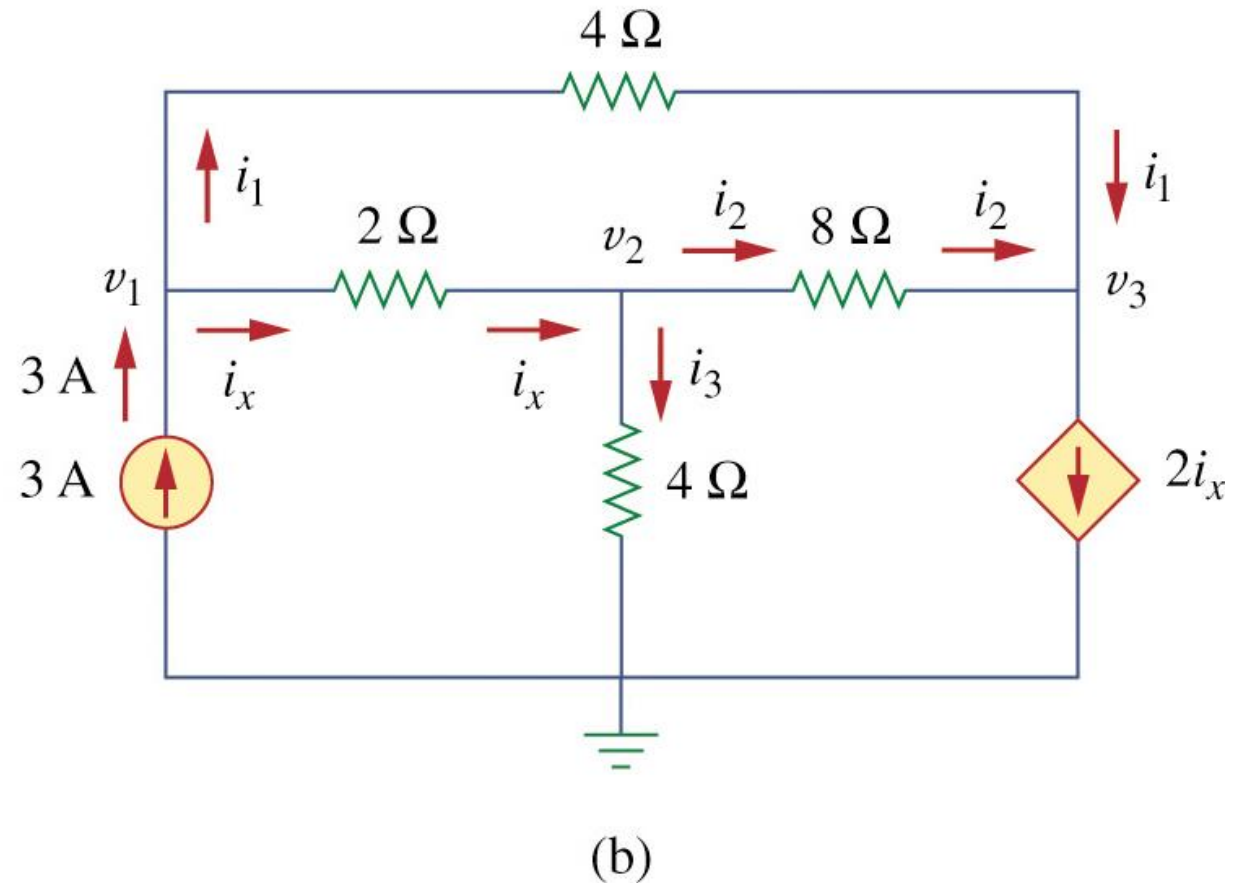


(b)

EXAMPLE 2:

- At node 1:

$$3 = i_1 + i_x$$
$$\Rightarrow 3 = \frac{v_1 - v_3}{4} + \frac{v_1 - v_2}{2}$$

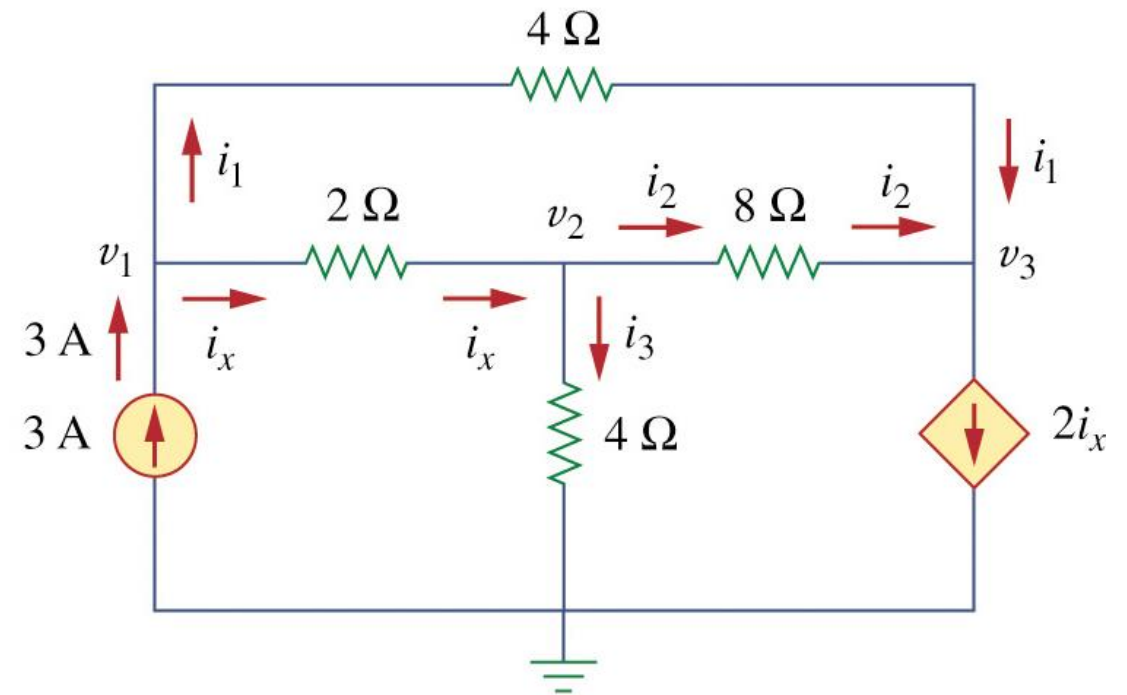


EXAMPLE 2:

▪ At node 2:

$$i_x = i_2 + i_3$$

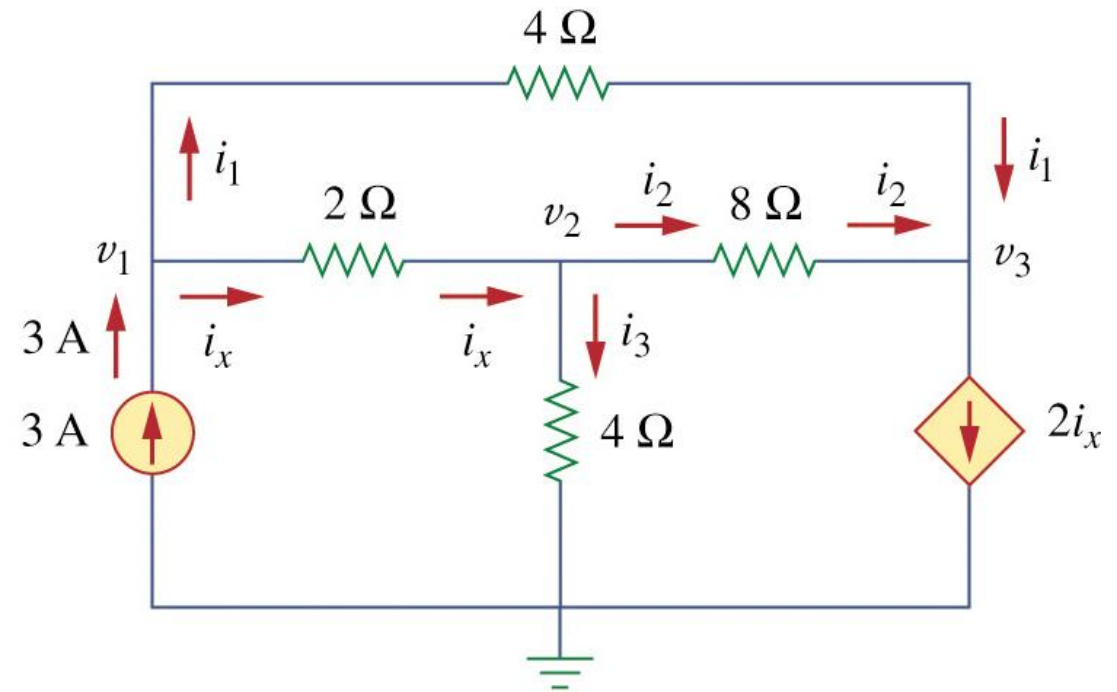
$$\Rightarrow \frac{v_1 - v_2}{2} = \frac{v_2 - v_3}{8} + \frac{v_2 - 0}{4}$$



EXAMPLE 2:

▪ At node 3:

$$i_1 + i_2 = 2i_x$$
$$\Rightarrow \frac{v_1 - v_3}{4} + \frac{v_2 - v_3}{8} = \frac{2(v_1 - v_2)}{2}$$



(b)

EXAMPLE 2:

- In matrix form:

$$\begin{bmatrix} \frac{3}{4} & -\frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{2} & \frac{7}{8} & -\frac{1}{8} \\ \frac{3}{4} & -\frac{9}{8} & \frac{3}{8} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$$

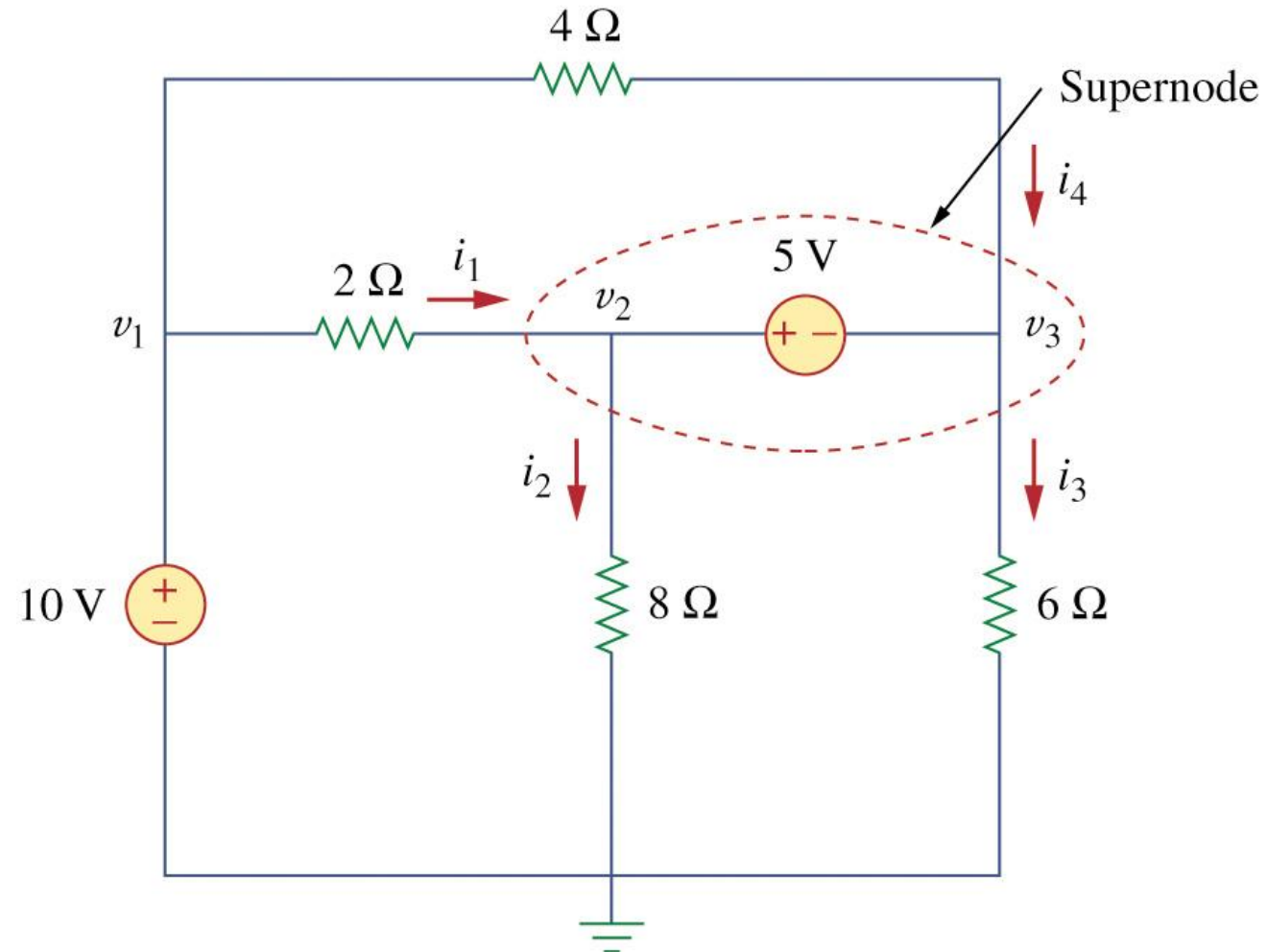
NODAL ANALYSIS WITH VOLTAGE SOURCES

- **Case 1:** The voltage source is connected **between a nonreference node and the reference node**: The nonreference node voltage is equal to the magnitude of voltage source.
- **Case 2:** The voltage source is connected **between two nonreferenced nodes**: a generalized node (supernode) is formed.



NODAL ANALYSIS WITH SUPERNODE

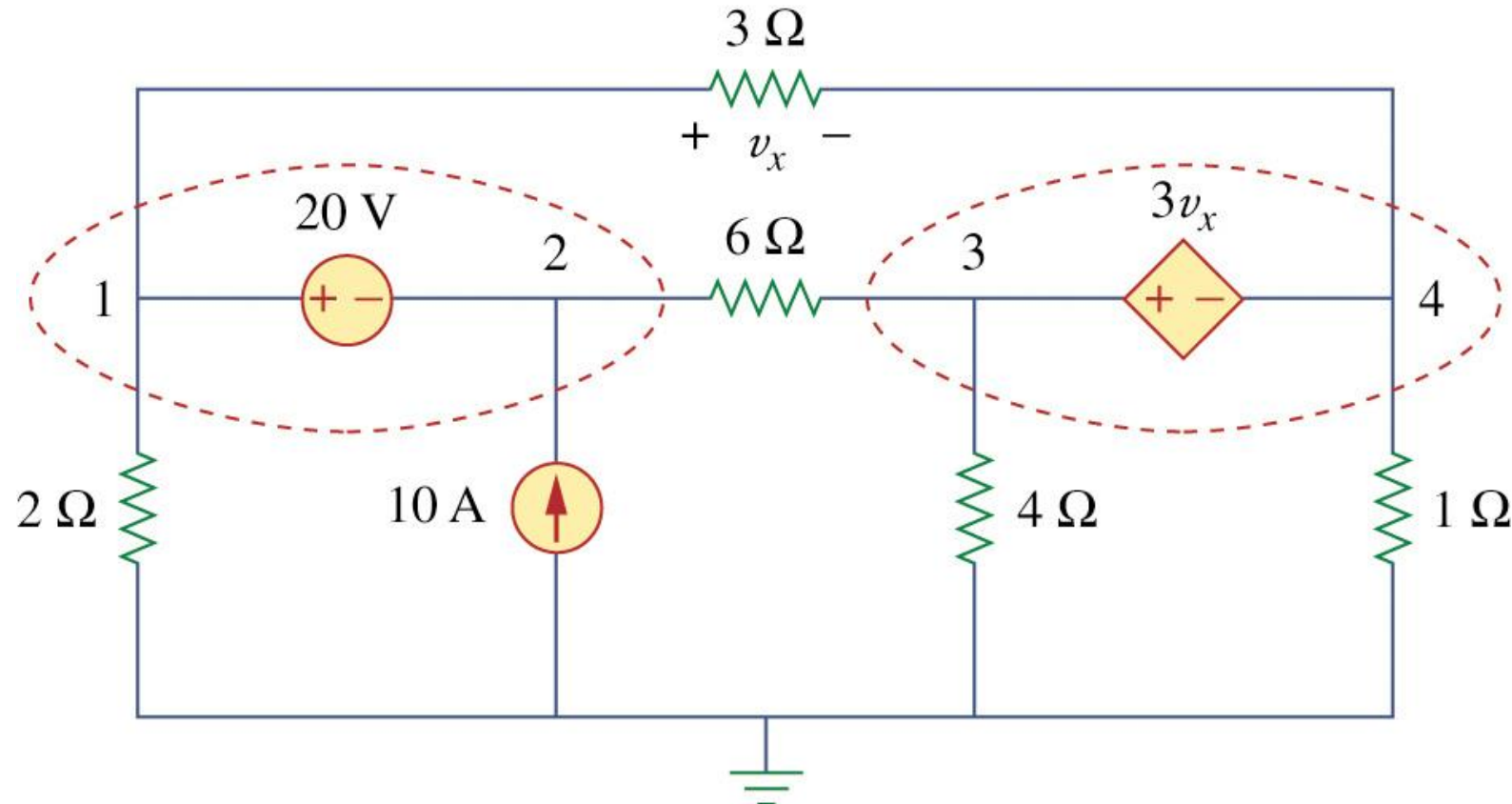
- A circuit with a supernode.



NODAL ANALYSIS WITH SUPERNODE

Example:

- Find the node voltages in the circuit below:

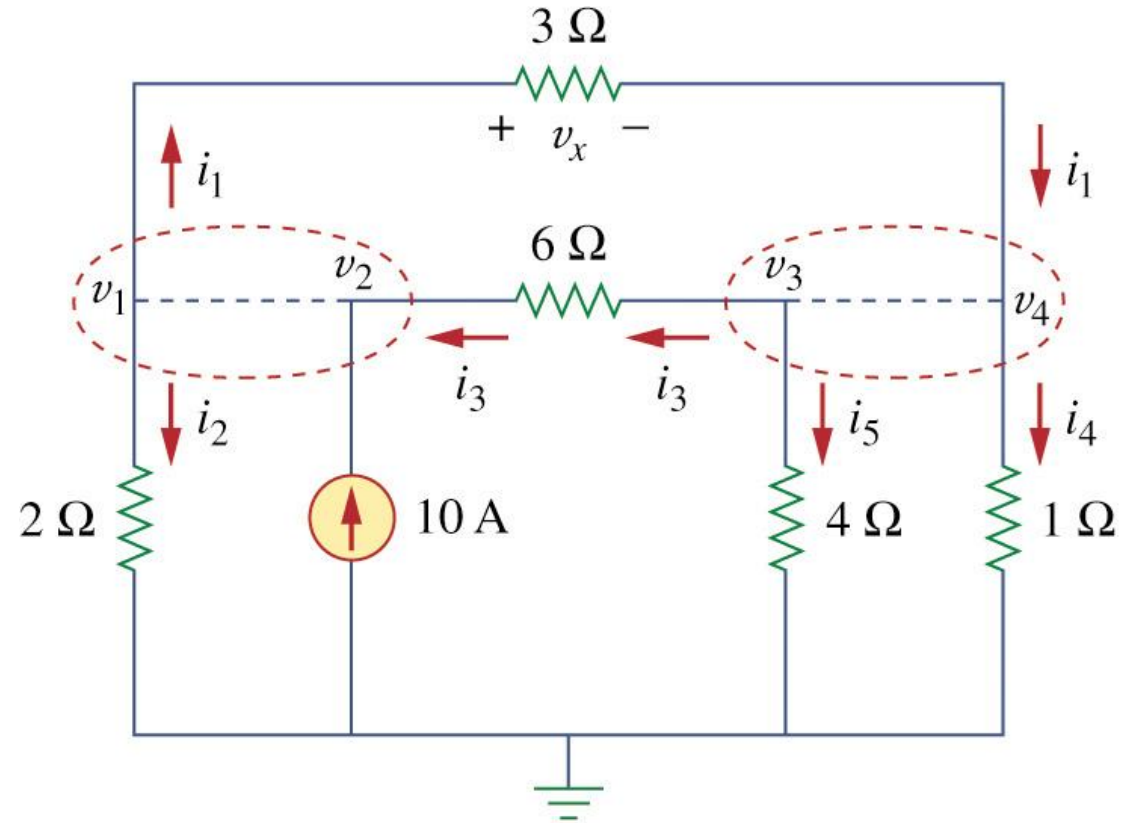


NODAL ANALYSIS WITH SUPERNODE

Example:

- At supernode 1-2:

$$\frac{v_3 - v_2}{6} + 10 = \frac{v_1 - v_4}{3} + \frac{v_1}{2}$$
$$v_1 - v_2 = 20$$



NODAL ANALYSIS WITH SUPERNODE

- At supernode 3-4:

$$\frac{v_1 - v_4}{3} = \frac{v_3 - v_2}{6} + \frac{v_4}{1} + \frac{v_3}{4}$$

$$v_3 - v_4 = 3(v_1 - v_4)$$

